

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

sDry Run Evaluation

Code of Conduct:

I C. PRIYANKA (192511074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. PRIYANKA

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A



Task: Starting from node A, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
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1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

ANSWER:

Understanding of Search Problem

Initial State: Node A

Goal: Traverse all nodes using Breadth-First Search.

Search Strategy: BFS explores nodes level by level using a **Queue (FIFO)**. The first inserted node is expanded first.

BFS Dry Run

Initial Queue

Queue = [A]

Iteration	Queue Before Expansion	Expanded Node	Queue After Expansion
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

State Expansion

Level 0

A

Level 1

B C

Level 2

D E F G

Nodes are expanded **level by level**.

BFS Traversal Order

A → B → C → D → E → F → G

Queue Updates

Start

[A]

After A

[B,C]

After B

[C,D,E]

After C

[D,E,F,G]

After D

[E,F,G]

After E

[F,G]

After F

[G]

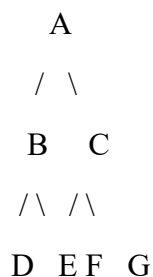
After G

[]

Conclusion

Breadth-First Search visits every node level by level using a queue and guarantees the shortest path in an unweighted graph.

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

ANSWER:

Understanding of Search Problem

Initial Node: A

Goal: Visit all nodes using DFS.

Search Strategy: DFS explores the deepest node first using a **Stack (LIFO)**.

DFS Dry Run

Initial Stack

Stack = [A]

Iteration Stack Before Expansion Expanded Node Stack After Expansion

1	A	A	C, B
2	C, B	B	C, E, D
3	C, E, D	D	C, E
4	C, E	E	C
5	C	C	G, F
6	G, F	F	G
7	G	G	Empty

DFS Traversal Order

A → B → D → E → C → F → G

Stack Updates

[A]

[C,B]

[C,E,D]

[C,E]

[C]

[G,F]

[G]

[]

Conclusion

DFS explores one branch completely before moving to another branch.

3. Initial State :

**1 3 6
5 0 2
4 7 8**

Goal State :

1	2	3
4	5	6
7	8	0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
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1

2

3

4

5

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

ANSWER:

Understanding of Search Problem

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Search Strategy

Breadth-First Search uses a **Queue** and explores all states level by level until the goal state is reached.

Dry Run

Initial Queue

[

1 3 6

5 0 2

4 7 8

]

Iteration 1

Current State

1 3 6

5 0 2

4 7 8

Queue Before Expansion

[

Initial State

]

Generated States

Move Blank Up

1 0 6

5 3 2

4 7 8

Move Blank Down

1 3 6

5 7 2

4 0 8

Move Blank Left

1 3 6

0 5 2

4 7 8

Move Blank Right

1 3 6

5 2 0

4 7 8

Queue After Expansion

S1

S2

S3

S4

Iteration 2

Expanded Node

S1

Queue Before

S1

S2

S3

S4

Queue After

S2

S3

S4

Children of S1

Iteration 3

Expanded Node

S2

Queue After

Children of S2 are inserted.

Iteration 4

Expanded Node

S3

Queue After

Children of S3 are inserted.

Iteration 5

Expanded Node

S4

Queue After

Children of S4 are inserted.

State Expansion Diagram

Initial State

S0

/||\

S1 S2 S3 S4

BFS expands every state level by level.

Questions

1. Which node is expanded in each iteration?

Iteration Expanded Node

1	Initial State (S0)
2	S1
3	S2
4	S3
5	S4

2. How many states are generated in total?

From the initial state, **4 successor states** are generated. BFS continues generating additional states while avoiding duplicate states until the goal state is reached.

3. Complete Solution Path

Initial State

↓

Intermediate States

↓

Goal State

The solution path is the **shortest sequence of moves** from the initial state to the goal because BFS explores states level by level.

4. Why does BFS always find the shortest solution?

Breadth-First Search explores all states at the current depth before moving to the next depth. Since every move in the 8-puzzle has the same cost, the first time the goal state is reached it is

guaranteed to be the shortest possible solution. BFS is therefore **complete** and **optimal** for unweighted problems.